

PERiFORM – WP3: Evaluating Methods for PPS Remapping in VR

Project Title: *Pushing boundaries: Peripersonal space as a tool to improve sports performance*

Funding Agency: Cyprus Research and Innovation Foundation (RIF) under the RESTART 2016–2020 framework (POST-DOC/0524/0011)

Work Package: WP3 – Evaluating Methods for PPS Remapping in VR

Lead Institution: University of Limassol

Principal Investigator: Dr. Christodoula K. Gavriel

Overview

WP3 investigates whether different types of movement — tool-based, upper-body, and whole-body — can reshape **peripersonal space (PPS)**, measured with a validated visuo-tactile detection task in VR.

Objectives

Validate and compare three VR-based PPS training interventions

Replicate the effectiveness of Pull Training (Petrizzo et al., 2024)

Explore whether torso-only or whole-body movement reshapes PPS without tools

Experimental Procedure

1. Intake

Participant arrives

Completes **informed consent** and **questionnaires**:

- ACS/ATTC <https://forms.microsoft.com/e/4aqnE5s5XT>
- ASRS <https://forms.microsoft.com/e/WnWJPtvsxN>
- TEQ <https://forms.microsoft.com/e/RBDmrUPfa9>
- SSQ <https://forms.microsoft.com/e/kccqDpiqdc>

Calibrates eye-tracking (must notify experimenter when calibration ends)

2. PPS Size Measurement Task

Run **four times total**:

Once **before any intervention** (PRE)

Once **after each intervention** (TOOL, TORSO, WHOLE)

Task details:

Before each block:

Enter subject ID, age, gender

Select Task 1 for the PPS task

Participant stands still in VR at a fixed point

Calibration step before each block

Participant follows a dot with their eyes in VR

Participant must notify experimenter when calibration is complete

After Calibration:

Start task by pressing Z key

Holds the **controller in right hand**, looks **into the eyes of the middle monster** at all times

Responds **only to vibrations**, not to approaching bubbles

3 trial types:

Bubble only

Vibration only

Bubble + vibration (majority of trials)

140 trials, ~12 minutes

Subject ID Format:

Example: 01POSTTOOL_CG_PERiFORM_WP3

01 = participant number

PRE = baseline

POST = after training

TOOL, TORSO, WHOLE = intervention type

CG_PERiFORM_WP3 = participant's initials + project + work package number

3. Training Interventions (VR Tasks)

Run in **counterbalanced order** with **15-minute breaks** between each (see file named "Counterbalanced order Exp1")

A. Pulling Task (Tool-Based)

Folder: Irene 5

Use VR shovel to **pull bubbles toward body** (participant stands inside target circle)

Bubbles may appear **left, center, or right**

Participant uses **controller in right hand**, secured with **wrist cord (κορδόνι)**

Duration: ~10 minutes

Select **Task 4** before starting

B. Torso Movement (Upper Body Only)

Folder: Christodoula - PERiFORM

Participant stands at the center of the lab, facing the same direction as in the other tasks, and leans torso side-to-side without stepping.

Follows visual torso body exercises presented by a virtual avatar.

Duration: ~10 minutes

During task:

Ask participant to **count repetitions out loud** from 1 to 10 for each movement

Monitor posture and accuracy of each exercise

Participant informs you when ready to move on

Press **N key** to advance to the next movement

⚠ If orientation is off in the VR scene, use the **first menu option** when launching the task to recenter the participant's position

C. Whole-Body Movement (Walking on Treadmill)

Folder: Christodoula - PERiFORM

Participant walks along a VR path while on a **real treadmill**

VR path duration: 10 minutes

Safety Protocol:

Attach **safety pin** to participant's shirt

Set **treadmill speed to 1.4**

Set **VR walking speed to 1.2**

Use the **last option in the launch window** to set VR walking speed

Participant holds **side handles (χειρολαβές)** throughout

Start treadmill first, then VR

Stop VR first, then treadmill

If participant's VR orientation is off, use the **first menu option** to fix their position

↻ After **every** VR task (including PPS), **fully close and relaunch the app**.

Participants Profile

N ≈ 34

Age: 18–40

Recruited via Sona System

Receive **course credit**

Session duration: ~2 hours

Data Analysis

Fit PPS boundaries using sigmoid function

Compute d' and c (Signal Detection Theory)

Two-way ANOVAs (Condition × Time)

Correlate pupil size with PPS boundary shifts

Task Controls

Task	Action	Key/Button
PPS Measurement	Start task	z key
PPS Measurement	Eye calibration	Follow dot in VR
Torso Task	Advance to next exercise	N key
All Tasks	Restart app between tasks	Manually quit app

Checklist for Each Participant Session

- ☐ Consent & questionnaires
- ☐ PPS Measurement Task (PRE)
- ☐ Training A → PPS (POST A)
- ☐ Training B → PPS (POST B)
- ☐ Training C → PPS (POST C)
- ☐ Relaunch app between each task
- ☐ Attach **safety pin** for treadmill
- ☐ Set speeds: **1.4 (treadmill), 1.2 (VR)**
- ☐ Launch treadmill before VR; stop VR before treadmill
- ☐ Use **first menu option** to fix VR orientation if needed
- ☐ Use **last menu option** to adjust VR walking speed if needed
- ☐ During torso task:

- ☐ Ask participant to count reps out loud (1–10)
- ☐ Monitor correct movement
- ☐ Press N to move to next exercise
- ☐ Save all data to PERiFORM Data/

Citation

Using methods or scripts for this work package:

Petrizzo, I., Mikellidou, K., Avraam, S., Avraamides, M., & Arrighi, R. (2024). Reshaping the peripersonal space in virtual reality. *Scientific Reports*, 14(1), 2438.